

Simplified 16 Line Message Format

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Background:

The traditional 16 line message is widely associated with radio teletype (RATT). Traditional RATT formats such as AC126/ACP-127 are discontinued or no longer maintained. ACP-128 is the last evolution of the format and is utilized with ALTERS. ACP126/127/128 are each described in their own book of unique, complex rules and are rich with anachronistic features.

This paper describes a generic, simplified, traditional 16 Line message format that retains basic addressing and message identification components while supporting preparation, transfer, printing and storage using personal computer and other microprocessor systems and technology. The format is simplified to the extent it can be explained on a single sheet of paper.

The Simplified 16 Line Format:

The simplified, generic 16 line resembles an abbreviated version of ACP128 and is not intended to process record traffic through ALTERS. It is suitable for radio communications on modern laptop computers utilizing low bandwidth HF/MF/LF circuits. This new format provides radio communications personnel a minimalistic and consistent approach to transmitting, receiving and relaying text messages while eliminating the complexity, redundancy and esoteric features of legacy formats.

The Simplified 16 Line format:

- Implements the ACP128 concept of a single Routing Line that is greatly simplified, may be used with either call signs or routing indicators, and contains all the components necessary to identify the message uniquely.
- Removes redundant and unnecessary features and rules:
 - a. Start and End of Message components are simplified.
 - b. The redundant security warnings, precedence, date and time, serial numbers, group counts, etc. are removed.
 - c. Addressees are simply TO. Eliminated INFO and EXEMPT addressee types.
 - d. Rules regarding capitalization, line length, page breaks, spacing are not included.
- Contains only the following elements:
 - a. FL1: Start of Message (VZCZC)
 - b. FL2/3: A Routing Line
 - c. FL4 optional: Handling instructions in plain language if required
 - d. FL6 (opt) FM followed by the Plain Language Address or name of the originator
 - e. FL7 (opt) TO followed by the Plain Language Address of name the Addressee
 - f. FL11: A separation (BT)
 - g. FL12: Content, such as code groups, free text, USMTF formatted text messages or data strips such as METAR or TAF
 - h. FL13: A separation (BT)
 - i. FL14 (opt) A hash or digest, other instruction/information about the message
 - j. FL16: End of Message without the 8 preceding blank lines (NNNN)

The abbreviated version of the format is illustrated in Figure 2 and demonstrates how 16 line format is used for radio communications such as sending data strips to a collection point or communications between radio stations in the same network. It requires the fewest keystrokes to draft and contains the minimal header information required to uniquely identify the message, and the sending and receiving station(s).

Figure 3 depicts the same message, with Precedent and Security components added. The codes for Precedent are :

- RR for Routine
- PP for Priority
- OO for Immediate
- ZZ for Flash

```
VZCZCMMM999
RTTUZHSW UHHX0063 2691810UUUU--UHHWH.
ZNR UUUUU
R 251810Z SEP 2020
FM USA HF GTWY SBMR HI
TO UHHWH/NETC G3-5 CUOPS HQ ARMY MARS FHU AZ
BT
UNCLAS|
this demonstrates am ACP-128 message
BT
#0063
```

NNNN

Figure 1- The ACP128 format

```
VZCZC
DESERTEAGLE DE GARDENRIDGE #0063 2691814
BT
this demonstrates a simplified 16 line message.
BT
```

NNNN|

Figure 2 – the 16 Line format that is simplified for Point to Point Radio Communications

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VZCZC
RR DESERTEAGLE DE GARDENRIDGE #0063 2691817
BT
UNCLAS
this demonstrates a simplified 16 line message.
BT
```

NNNN

Figure 3 – the 16 Line format, abbreviated format with precedence and security elements.

Drafting A Message in 16L:

The Start of Message provides a unique symbol to start processing the following text. Simply put VZCZC on its own line.

The FL2/3 Routing Line consists of the following elements, listed from left to right and separated by spaces. Place these all on one line.

- The Precedence: RR, PP, OO or ZZ.
- The Call Sign (s) or Routing Indicator(s) of the called or destination station(s). Multiples are separated by a space. Either call signs or routing indicators may be used but not together in the same message. See Figure 3 and 4 for examples using both call-signs and routing indicators.
- The pro-sign DE which means "From".
- The Call Sign or Routing Indicator of the sending or originating station. Either call signs or routing indicators may be used, but not together in the same message. See Figure 3 and 4 for examples using both call-signs and routing indicators.
- The serial number in four digits with leading zeroes and prefaced with a #
- The Julian Date and Time in the following format: JJJHHMM

```
VZCZC
RR UHWHH DE UHHX #0063 2691817
BT
UNCLAS
this demonstrates a simplified 16 line message.
BT

NNNN
```

Figure 4 – The generic 16 Line, abbreviated format utilizing Routing Indicators

The Routing Line is followed by an optional, single line of text for routing and handling instructions between radio operators using plain language or pro-signs. Instructions such as RELAY TO, REPEAT BACK, DO NOT ANSWER, etc may be placed in this line as required. This line is contextual between radio operators and should be deleted when the instruction is no longer relevant or before final delivery.

A full "PLAINDRESS" message is illustrated in Figure 5. This is the highest level of verbosity and includes traditional header elements of FL2, 3 and 4 along with addressing elements associated with FL6, and 7. Add originator and each addressee on their own lines if required.

```
VZCZC
RR DESERTEAGLE DE GARDENRIDGE #0063 2691817
FM USA HF GTWY SBMR HI
TO UHWHH/NETC G3-5 CUOPS HQ ARMY MARS FHU AZ
BT
UNCLAS
this demonstrates a simplified 16 line message.
BT

NNNN
```

Figure 5 - The 16 line message, in full "PLAINDRESS" format. Either Routing Indicators or Call Signs use the same generic message format.

Separate the header from the message body with a "break". Put "BT" on its own line.

A security warning may be placed at the first line of the message text if it is required. Use common security warnings that are immediately understood by personnel and approved in your organization, such as UNCLASS, UNCLASSIFIED, CUI, SECRET, etc.

The actual message body comes next. It may be a USMTF format, text data strips such as METAR or TAF strips, or free form text that expresses the thoughts and ideas of the author. The message body may be as long as is reasonable for the radio circuit, without page breaks. Both upper and lower case text is permitted, and TYPING IN ALL CAPITAL LETTERS IS DISCORAGED. Line length is according to the text or data strip format utilized.

When code groups are sent as the message body, any of the abbreviated variations depicted in Figures 2, 3 or 4 may be utilized as appropriate. The PLAINDRESS format with FM and TO lines as shown in Figure 5 is never used with code groups.

Separate the message body from the ending of the message. Put BT on its own line.

The last line before the End Of Message is an optional set of instructions or information relating to the message content. This line is between the originator and recipient and may contain whatever plain language, hash codes, or other information that is relevant to the parties addressed in the message.

The message ends with the traditional NNNN, however the 8 preceding blank lines that were intended for advancing tractor feed paper from a printer are no longer necessary and are eliminated.